ISO 9001
ISO 14001
OHSAS 18001

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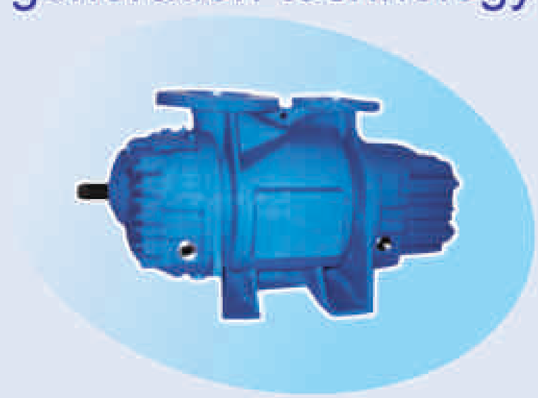
ROTARY VACUUM BLOWERS
WITH AIR INJECTION

......the next generation technology



Unmatched Performance
Superior Technology
Trouble Free Operation

Bulletin No. SP-012

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THE COMPANY

Swam a leader in Rotary Blower technology, has during its nearly three decades of experience in design & manufacturing of blowers has acquired exceptional product development capabilities to meet the stringent quality and performance requirements and to conform to the customer's tough & diversified specification. Emphasis has been on innovative design, precision engineering and responsiveness to customers.

Swam, thus offers its advance technology next generation **Rotary Dry Vacuum Blowers;** equipped with latest features, enabling it more reliable, energy efficient & lower maintenance.

Blowers are ruggedly built with appropriate rotor shaft design, aiming at low deflection & critical consideration. The rotors are supported on heavy duty antifriction oil lubricated bearings. The rotor profile of vacuum blowers are computer generated enabling better uniform clearances and higher volumetric efficience.

The Swam blowers have very high accuracies & manufactured on CNC machines. The Rotary Dry Vacuum Blowers are available for flows upto 65000 m^3/hr. and pressure upto 1.10 kg /cm^2 in single stage. The blowers are suitable for continuous operation to work as compressors or as exhausters.

FOCUS ON QUALITY & RELIABILITY

Emphasis on quality & performance is a commitment of every individual at **SWAM** beginning at grass root level.

The quality control is practiced at all levels of the organization and all its aspects are thoroughly complied with the specified standard and specification, till the product gets delivered.

The quality assurance system encompasses self quality control by each worker, supervised by quality control engineers. Each blower undergoes thorough & stringent tests at different stages of manufacturing and are test run for several hours prior to despatch as per the relevant code, customers' specification & approved quality plan.

These functions are carried out by our team of dedicated, qualified & experienced engineers.

The company's quality system is as per ISO 9001 : 2008 and has very high regard for safety & environment and hence has it's all plants ISO 14001 : 2004 and OHSAS 18001 : 2007 certified.

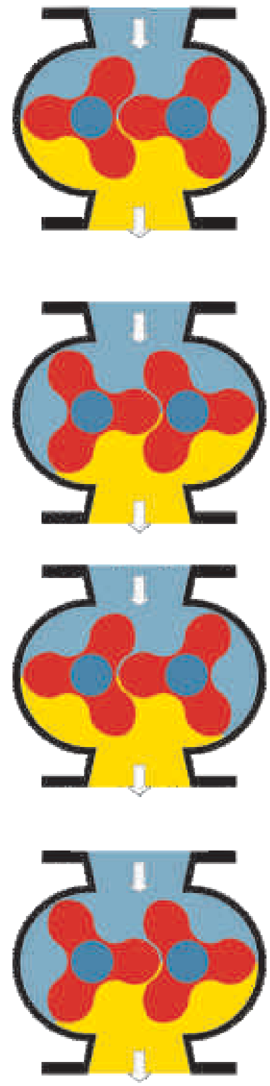
ADVANCE MANUFACTURING SYSTEM

Swam has the most advance manufacturing system with a large number of CNC Machines installed in our four plants. The blowers and every accessory are manufactured with most optimum profile resulting higher efficiencies, high displacement, lower slippage. Thus these are more energy efficient blowers compared to any other.

WORKING PRINCIPLE

The working principle of Tri-lobe blowers is similar to that of Twin-lobe type. The rotors, move opposite each other, maintaining fine clearances, in perfect synchronization through a set of precision timing gears.

As the rotation proceeds, the trapped gas moves along until it reaches the discharge port raising its pressure against the system resistance. The precise clearances within the blower-makes possible the rise in pressure with least internal slip loss. The delivery is non-pulsating smooth and with lower sound emission.



AXCEL SERIES

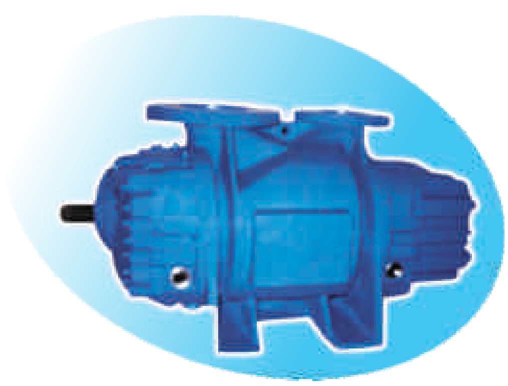
The next generation Axcel Series Blowers offers unmatched performance operation and negligible maintenance due to the superior of advance features.

The blowers are manufactured on sophisticated advance CNC machines and the rotor profiles are levy pre-lene and accurate.

The Axcel Series are latest in technology in the Rotary Blower Design and are energy efficient with guaranteed negligible maintenance. These blowers have features, that make them unmatched in performance and high energy efficient.

Salient Features

- Ruggedly built for working under tough conditions, high load transmission and higher operating speeds.
- Computer generated Axcel Series lobe-profile optimizes volumetric efficiency.
- Helical heavy duty timing-gears, shrunk fit on rotor shafts make reliable and accurate fitment.
- Guaranteed **oil free conveyance.**
- Effective oil lubrication of at both end bearings and gears.
- Suitable for positive and negative conveying and for top or bottom suction.
- Side suction or side discharge arrangement available on select models.
- Special sealed models for gas conveying and for vacuum application.
- Provided with pulsation Dampeners.
- Rotating parts dynamically balanced to finer grades.
- Six number of bearings as minimum.
- Heavy Duty Shaft.
- Special Double Acting Metallic Seals.



28000 cmh Blower for Steel/Metallurgical Industry

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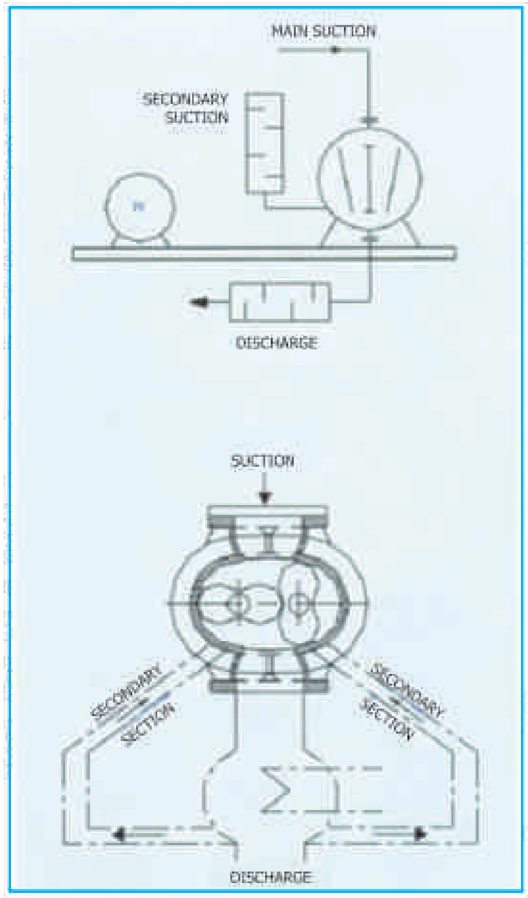
The conventional Rotary Vacuum Blowers working, as exhausters do not develop vacuum exceeding 550mbar. This limitation is due fo the temperature rise within the blower. Thus for achieving higher vacuum, the choice falls either on Water Ring or Rotary Vane type vacuum pump.

Swan Rotary Vacuum Blower MHV - Series have overcome this constraint. The blower has secondary suction with help develop vacuum upto 800 mbar. The blower has completely dry operation and does not require any sealing fluid.

The pumps are reliable for continuous operation, require lower maintenance and consume less power compared to other mechanical pumps.

WORKING PRINCIPLE

The Swam blower works on the principle of Rotary (Roots Type) blowers wherein two interlocked and synchronized parallel rotors, rotate in opposite direction within the compression chamber, without mechanical contact, but maintaining very fine clearances of tens of microns. Through its secondary suction, cool atmospheric air is sucked and reduces the temperature of the hot air. The permits further rise of vacuum within the pump chamber.



DESIGN AND CONSTRUCTION

The uniqueness of the design lies in its secondary suction, which is an integral part of the blower. Through this the cooling media enters the compression chamber, adequate to keep discharge air temp. within 135°C even at high vacuum of 800 mbar.

In case of air service blowers, the cooling media is atmospheric air but for gas pumps, it is the discharge gas routed through a cooler. No other cooling fluid required.

PERFORMANCE

The flow ratings of the standard models are listed in the performance table along with the other technical details like pump speed, bkw etc. The maximum allowable vacuum at operating speed is also indicated therein. The performance testing is carried out as per BS-1571 Part II. The volume flow 'Q' is in m^3/hr. at suction conditions.

WITH AIR INJECTION

PERFORMANCE TABLE

VACUUM in mbr		500		600		700		800	
MODEL	SPEED	Q	BKW	Q	BKW	Q	BKW	Q	BKW
40 MHV	2985	1806	30.6	1788	37.4	1763	42.9	1745	48.5
	2060	1170	20.7	1158	24.6	1134	28.6	1115	32.4
	1460	768	14.4	750	17.3	726	20.1	708	22.9
75 MHV	2250	2065	35.8	2030	43.0	2005	50.0	1975	58.2
	1800	1638	28.9	1608	34.5	1581	40.0	1554	45.7
	1450	1271	21.6	1241	26.0	1214	30.0	-	-
116 MHV	1800	3275	56.4	3220	67.6	3166	77.9	3108	88.2
	1450	2630	45.5	2582	54.4	2538	62.8	2492	71.1
	1200	2090	36.2	2043	44.0	1998	51.0	1952	58.0
170 MHV	1480	6653	113.4	6538	135.0	6423	156.7	6308	179.0
	1200	5174	90.7	5059	108.2	4959	125.4	4869	143.0
	985	4137	75.0	4022	89.0	3922	103.0	3822	117.0
235 MHV	1350	12000	202.0	11795	239.0	11665	277.0	11565	314.0
	1150	10210	172.0	10040	203.0	9930	231.0	9850	266.0
	985	8460	143.0	828	170.0	8160	198.0	8040	225.0
321 MHV	985	14680	272	14435	323	14200	374	13965	425
	750	12360	227	12115	269	11880	311	11645	354
	740	10514	200	10269	236	10034	271	9799	400

APPLICATIONS

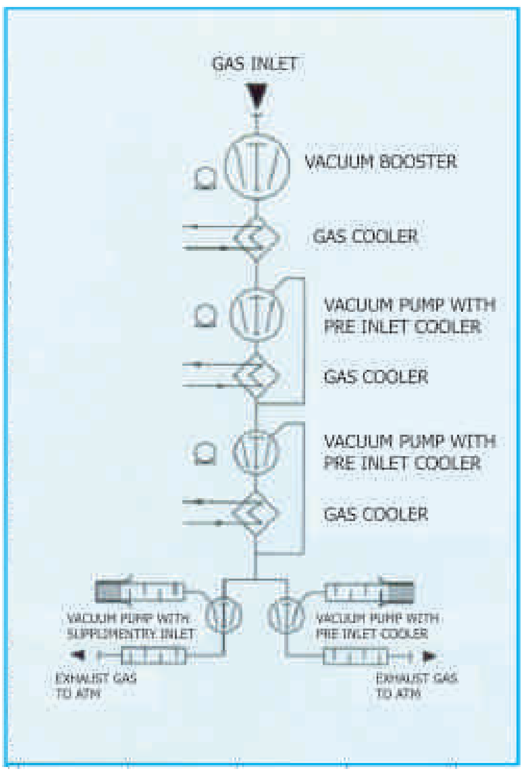
The MHV series pumps are ideally suited for vacuum upto 800mbar and offer better alternate to Water-ring or Rotary Vane type pumps. The pump absorbs less power.

Fewer parts are subject to wear & hence maintenance is also much less. Since there is no sealing fluid, the compression is absolutely dry. This feature makes pump suitable for handling any type of neutral gas, and air handling pump can discharge direct into atmosphere without separate being installed.

The pumps are suited for a large number of modern vacuum applications and processes such as pneumatic conveying under negative pressure, exhausting and recirculating of gases for processes in pharmaceutical, food, chemical, cement, and metallurgical industries.

SPECIAL PACKAGE

Multistage pumps can be packaged for obtaining high vacuum of below one mbar. The schematic layout may be referred. The arrangement is most suited for high displacement & faster evacuation, such as in VODC system of stainless and alloy steel plants and similar other applications.



Schematic Layout of Multi Stage Vacuum System

ROTARY DRY VACUUM BLOWERS

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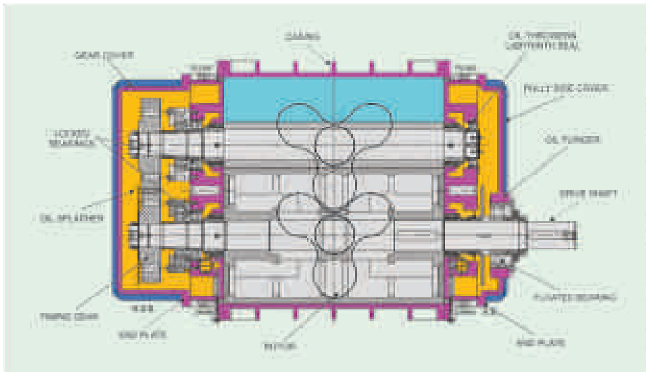
CONSTRUCTION



Casing

It is a one piece Integral Grey or Ductile Cast Iron or Cast Steel casting.

The surface of casing is adequately ribbed for strengthening against distortion on high loads. Selective models have sound attenuating chambers on discharge. The suction flange is on the top and discharge is either at bottom or on the side.

Sealing

The compression chamber is sealed from the oil chambers through effective labyrinth seal provided with a neutral chamber and the air conveyed is guaranteed oil free. The drive shaft end has a lip sealriding on hardened and replaceable sleeve.

For conveying gases the seal can be mechanical (M) or gland packed type (P).

Lubrication

There is an effective OIL splash lubrication for gears and bearings. This arrangement is for all blowers. However, for certain specific applications the blowers are provided with forced lubrication (L). For blowers operating at low temperature heating arrangement is provided.

Timing Gears

The timing gears are hardened and ground, heavy-duty helical design to with-stand wear. Fixing to the shaft is through press on tapers under high oil pressure.

Rotors

Each rotor has 3-lobes of computer generated profile optimizing volumetric efficiency.

The lobes are generally of fine grain Grey Cast or Ductile Iron and has pressed in steel shafts. The lobe materials can also be SS or AL depending on the process requirement. The ends of lobes are plugged; when handling dust laden air.

The rotors of smaller blowers can be out of forged steel with integral shaft. The rotors are dynamically balanced to very fine grade of 2.5 of ISO-1940.

Rotors pair for 15000 cmh blower

STANDARD EXECUTION

The standard units consists of :

- Fabricated steel base frame with C.I. slide rails
- Non-return valve
- Discharge silencer
- V-belts and pulleys or coupling with guard
- Vacuum gauge
- Flexi-mounts
- Suction strainer
- Silencers on main and secondary suction

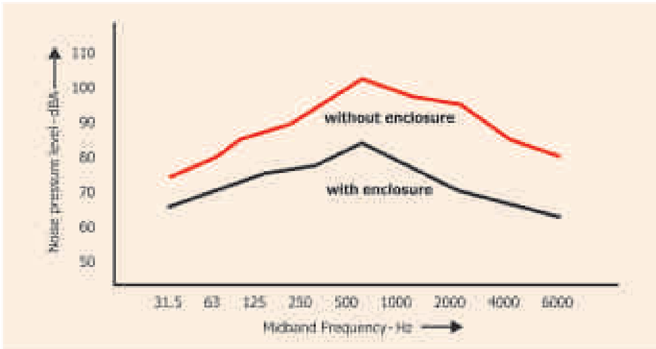
OPTIONAL

- Temperature gauge and switch
- Vacuum gauge and switch
- Flexible belows
- Cooler on discharge
- Acoustic enclosure for sound reduction

SALIENT FEATURES

- **High volumetric and mechanical efficiencies**
- **Lower power consumption**
- **Lower operating costs**
- **Absolutely dry operation**
- **Guaranteed oil free conveying**
- **Reliable operation and less maintenance**

NOISE CHARATERISTIC



The noise characteristic of a specific model of 24500 M^3/W at a gauge pressure of 0.7 kg/ cm^2g operating at 1000 rpm.

The rotary machines generate high pitch sound due to its operational characteristic. This, however, depends on operating speed, differential pressure and on rotor geometry.

It can be reduced by using absorptive or reactive silencers, at the suction and discharge of the blower.

There is, however, limitation to the sound reduction by mere use of silencers. Sound proof acoustic enclosures are used for further reduction upto 25 dBA.

AIR FLOW REGULATION AT CONSTANT DISCHARGE PRESSURE

For regulating the flow of air at a constant discharge pressure either of the following methods can be adopted:

- Blower operates at constant speed and the excess air vented or bye-passed through a manually operated or an automatic pneumatic operated valve.
- By regulating blower speed through a variable frequency drive or by using a dual speed motor.

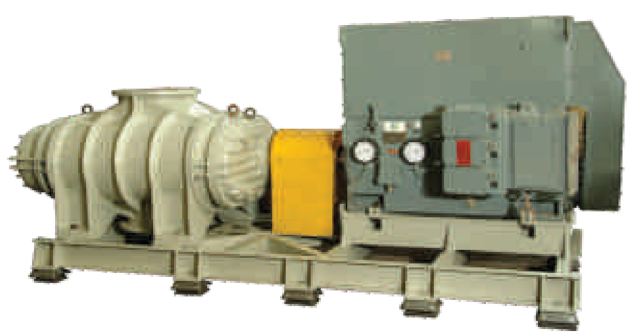
By 1st method the energy absorbed by the blower remains constant, where as in 2nd method it reduces in proportion to the air delivered.

Performance Testing

The performance testing of blowers is carried out for several hours at the rated conditions at the shop prior to dispatch as per the standard test procedure. Swam has the biggest shop test facilities with testing up to 1500 HP with voltages 415 V, 690 V & 6.6 kv / 6600 V.

During testing the various aspects are checked, such as rated flow, discharge pressure, absorbed power, vibration, noise and outlet temperature etc. as per the approved quality plan.

Each blower dispatched, carries a FACTORY TEST REPORT and a Guarantee Certificate.



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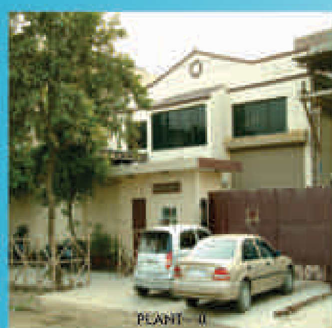
Customer Support Services:

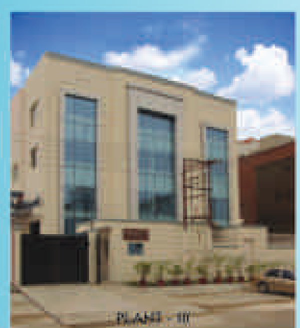
A team of experienced and qualified engineers are deployed to rendering all kind of technical post supply services to our esteemed customers; such as pre-commissioning and post commissioning services, trouble shooting etc. etc., to make the equipment to user's satisfaction.

Professionally & technically competent personnel are always ready to render these services, promptly and expeditiously.

For details contact H.O. or the nearest service network.

Manufacturing Plants:





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SWAM PNEUMATICS PVT. LTD.

(ISO - 9001:2008 Quality Certified)

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For details of our local contact office in your region, contact H.O. or log on to website

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